



Integrantes

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Laboratório 3 - Redes de Comunicação I

1 – Tabela de Hosts e Redes

LAN 1

PC1	192.168.10.11
PC2	192.168.10.12
PC3	192.168.10.13
PC4	192.168.10.14
Default Gateway	192.168.10.1

LAN 2

PC5	192.168.20.15
PC6	192.168.20.16
Default Gateway	192.168.20.1

LAN 3

PC7	192.168.30.17
PC8	192.168.30.18
Default Gateway	192.168.30.1

2 – Tabela de Enlaces entre Roteadores

Enlace/Roteadores Ligados	Rede
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R1 ↔ R2	10.0.12.0/30
R2 ↔ R3	10.0.23.0/30

Router 1

Interface	Endereço
g0/0	192.168.10.1/24
g0/1	10.0.12.1/30

Router 2

Interface	Endereço
g0/0	192.168.20.1/24
g0/1	10.0.12.2/30
g0/2	10.0.23.1/30

Router 3

Interface	Endereço
g0/0	192.168.30.1/24
g0/1	10.0.23.2/30

3 – Saídas show ip interface brief e show ip route



Router 1

```
Router>show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.10.1    YES manual up          up
GigabitEthernet0/1       10.0.12.1       YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
```

```
Router>show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/1
L    10.0.12.1/32 is directly connected, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0
S    192.168.20.0/24 [1/0] via 10.0.12.2
S    192.168.30.0/24 [1/0] via 10.0.12.2
```

Router 2

```
Router>show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.20.1    YES manual up          up
GigabitEthernet0/1       10.0.12.2       YES manual up          up
GigabitEthernet0/2       10.0.23.1       YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
```

```
Router>show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/1
L    10.0.12.2/32 is directly connected, GigabitEthernet0/1
C    10.0.23.0/30 is directly connected, GigabitEthernet0/2
L    10.0.23.1/32 is directly connected, GigabitEthernet0/2
S    192.168.10.0/24 [1/0] via 10.0.12.1
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0
L    192.168.20.1/32 is directly connected, GigabitEthernet0/0
S    192.168.30.0/24 [1/0] via 10.0.23.2
```



Router 3

```
Router>show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.30.1    YES manual up          up
GigabitEthernet0/1       10.0.23.2       YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
```

```
Router>show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.23.0/30 is directly connected, GigabitEthernet0/1
L       10.0.23.2/32 is directly connected, GigabitEthernet0/1
S       192.168.10.0/24 [1/0] via 10.0.23.1
S       192.168.20.0/24 [1/0] via 10.0.23.1
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0
```

4 – Resultados de ping e tracert/tracert na mesma LAN

PC1 (LAN 1) -> PC2 (LAN 1)

```
C:\>ping 192.168.10.12
```

```
Pinging 192.168.10.12 with 32 bytes of data:
```

```
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.10.12:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>tracert 192.168.10.12
```

```
Tracing route to 192.168.10.12 over a maximum of 30 hops:
```

```
  0  0 ms    0 ms    1 ms    192.168.10.12
```

```
Trace complete.
```



5 – Resultados de ping e tracert/traceroute em LANs diferentes

PC1 (LAN 1) -> PC7 (LAN 3)

```
C:\>ping 192.168.30.17

Pinging 192.168.30.17 with 32 bytes of data:

Reply from 192.168.30.17: bytes=32 time<1ms TTL=125
Reply from 192.168.30.17: bytes=32 time=1ms TTL=125
Reply from 192.168.30.17: bytes=32 time=1ms TTL=125
Reply from 192.168.30.17: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.30.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

```
C:\>tracert 192.168.30.17

Tracing route to 192.168.30.17 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.10.1
  2  1 ms    0 ms    5 ms    10.0.12.2
  3  1 ms    0 ms    0 ms    10.0.23.2
  4  0 ms    0 ms   10 ms    192.168.30.17

Trace complete.
```

PC1 (LAN 1) -> PC5 (LAN 2)

```
C:\>ping 192.168.20.15

Pinging 192.168.20.15 with 32 bytes of data:

Reply from 192.168.20.15: bytes=32 time<1ms TTL=126
Reply from 192.168.20.15: bytes=32 time=1ms TTL=126
Reply from 192.168.20.15: bytes=32 time=1ms TTL=126
Reply from 192.168.20.15: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>tracert 192.168.20.15

Tracing route to 192.168.20.15 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.10.1
  2  0 ms    0 ms    0 ms    10.0.12.2
  3  0 ms    0 ms    0 ms    192.168.20.15

Trace complete.
```

PC5 (LAN 2) -> PC7 (LAN 3)

```
Pinging 192.168.30.17 with 32 bytes of data:

Reply from 192.168.30.17: bytes=32 time<1ms TTL=126
Reply from 192.168.30.17: bytes=32 time=1ms TTL=126
Reply from 192.168.30.17: bytes=32 time<1ms TTL=126
Reply from 192.168.30.17: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.30.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 192.168.30.17

Tracing route to 192.168.30.17 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.20.1
  2  0 ms    0 ms    0 ms    10.0.23.2
  3  0 ms    0 ms    1 ms    192.168.30.17

Trace complete.
```

6 – Breve explicação sobre roteamento estático

O roteamento estático permite a comunicação entre as três redes porque são configuradas rotas manualmente no roteador, indicando o caminho que os pacotes devem seguir para alcançar redes diferentes. Assim, quando um dispositivo envia um pacote para outra rede, ele encaminha ao gateway padrão (roteador), que, com base nas rotas configuradas, direciona o pacote para a interface correta até chegar ao destino.